Daily Logic Drill #2

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Necessary vs Sufficient Conditions · "Iff" · Chains of Implication · Weakest/Strongest Condition.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

Translate familiar if–then facts into necessary/sufficient language. Pure recognition.

1. Is "n is divisible by 4" a sufficient condition for "n is divisible by 2"?

2. Is "n is divisible by 4" a necessary condition for "n is divisible by 2"?

3. Is "$x = 3$" a sufficient condition for "$x^2 = 9$"?

4. Is "$x = 3$" a necessary condition for "$x^2 = 9$"?

5. True or false: "being a rectangle" is a necessary condition for "being a square".

6. True or false: "being a rectangle" is a sufficient condition for "being a square".

7. Is "$n = 6$" a sufficient condition for "n is a multiple of 3"?

8. Is "$n = 6$" a necessary condition for "n is a multiple of 3"?

9. True or false: "n is even" is a necessary condition for "n is divisible by 4".

10. True or false: "n is even" is a sufficient condition for "n is divisible by 4".

Section B Manipulation Drills (≤50 s each)

Classify a condition as necessary, sufficient, both, or neither; chain implications together.

1. Classify "$x > 5$" as a condition for "$x > 0$": necessary, sufficient, both, or neither?

2. Classify "n is divisible by 6" as a condition for "n is divisible by 2 and by 3".

3. Classify "$ab = 0$" as a condition for "$a = 0$" (where a, b are real numbers).

4. Classify "n^2 is even" as a condition for "n is even".

5. Classify "x is an integer" as a condition for "x is a rational number".

6. Classify "a triangle is equilateral" as a condition for "a triangle is isosceles".

7. Given that "$P \Rightarrow Q$" and "$Q \Rightarrow R$" are both true, is P a sufficient condition for R? Justify briefly.

8. Given that "n divisible by 9 $\Rightarrow$ n divisible by 3" and "n divisible by 3 $\Rightarrow$ digit sum divisible by 3", is "n divisible by 9" sufficient for "digit sum divisible by 3"?

9. Each of (a) n divisible by 8, (b) n divisible by 4, (c) n divisible by 2 is sufficient for "n is even". Which is the *weakest* (least restrictive) of the three?

10. Each of (a) n divisible by 2, (b) n divisible by 4, (c) n divisible by 8 is necessary for "n is divisible by 24". Which is the *strongest* (most restrictive) of the three?

Section C Substitution & Structure (≤90 s each)

Full necessary-and-sufficient proofs, implication chains, and your first four-way classification.

1. Which of the following best describes "n is a multiple of 6" as a condition for "n is a multiple of 2 and a multiple of 3"?

A) Necessary but not sufficient.

B) Sufficient but not necessary.

C) Necessary and sufficient.

D) Neither necessary nor sufficient.

2. Prove that "n is divisible by 4" is necessary and sufficient for "n is even and $n/2$ is even".

3. Classify "$x^2 < 4$" as a condition for "$x < 2$" (real x). Justify both directions.

4. Classify "n is prime" as a condition for "n has exactly two positive divisors". Justify your answer.

5. Given that P is sufficient for Q, and Q is sufficient for R, prove that P is sufficient for R.

6. Is "n is a multiple of 4" necessary for "n is a multiple of 12"? Is it sufficient? Justify both answers.

7. Find a condition C on a positive integer n that is necessary but not sufficient for "n is a multiple of 100". Justify your choice.

8. Given that x is necessary and sufficient for y, and y is necessary and sufficient for z, prove that x is necessary and sufficient for z.

Section D Challenge (TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. For real x, let P be "$x > 0$" and Q be "$x^2 > 0$". Is P necessary and sufficient for Q, necessary but not sufficient, sufficient but not necessary, or neither?

A) Necessary and sufficient.

B) Necessary but not sufficient.

C) Sufficient but not necessary.

D) Neither necessary nor sufficient.

2. Classify "x is rational" as a condition for "x^2 is rational" (real x). Prove both parts of your classification.

3. A year Y is a leap year exactly when Y is divisible by 4, except that century years (multiples of 100) must be divisible by 400. Is "Y is divisible by 4" necessary and sufficient, necessary but not sufficient, sufficient but not necessary, or neither, for "Y is a leap year"? Justify with a specific year.

4. Statements A, B, C satisfy $A \Rightarrow B$, $B \Rightarrow C$, and $C \Rightarrow A$. Prove that A, B, C are all logically equivalent (i.e. each pair is an "iff").

5. Statement S is: "If n is a multiple of 9, then n is a multiple of 3." Is "n is a multiple of 9" necessary and sufficient for "n is a multiple of 3"? If not, precisely which of necessary/sufficient does it satisfy, and what does S's converse (from Day 1's toolkit) have to do with your answer?

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."
